

Auction for Prominence and Price-Directed Search

Theory and Experiment

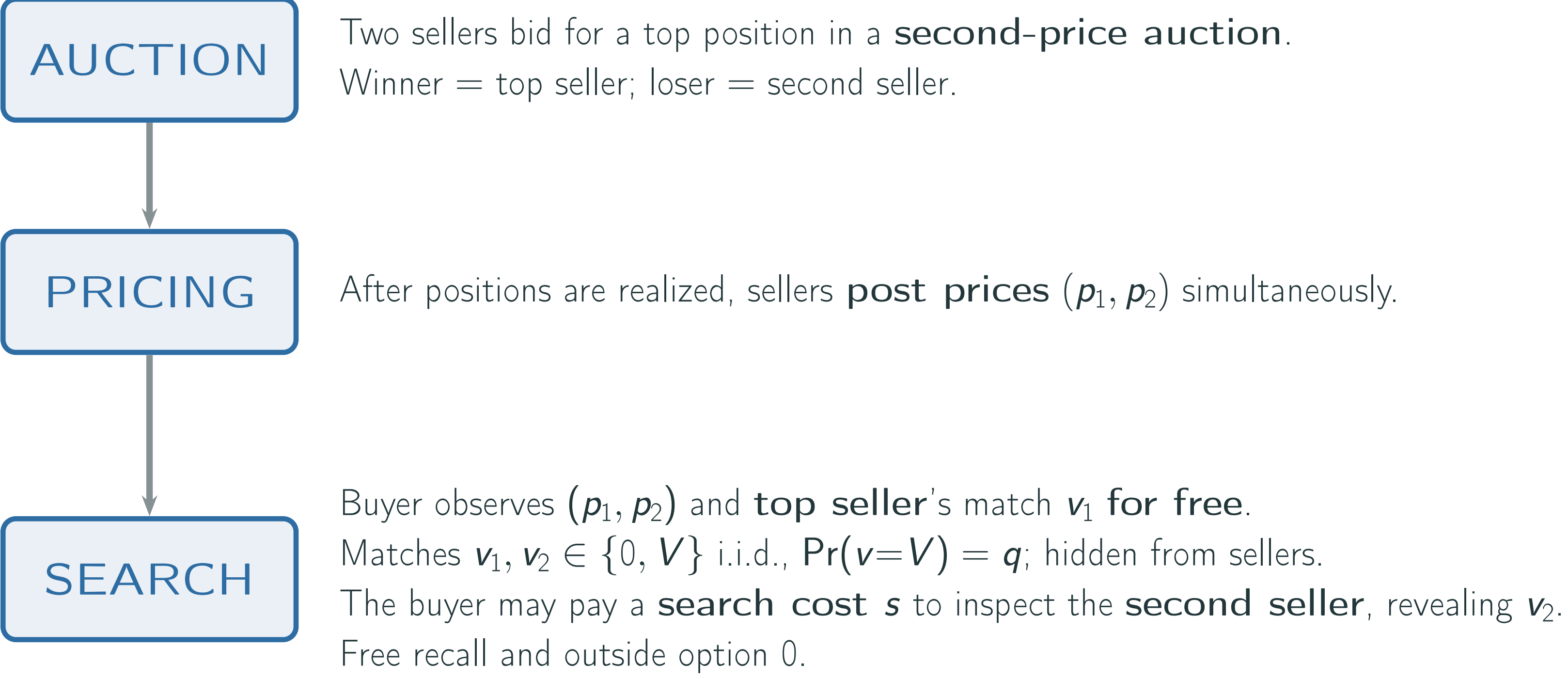
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A MARKET FOR SEARCH PROMINENCE

On digital marketplaces, search is costly, so sellers **pay to be inspected first** — Google search ads, Amazon sponsored products, Expedia featured placement.

- **Research question:** When sellers compete for prominence, how do prices and search behavior respond to the **magnitude** of the search friction? More broadly, should prominence be allocated through markets?
- **Could help:** markets and prices may sort buyers to better-matching products.
- **Could hurt:** money-burning and rent-seeking.
- **This paper:** a parsimonious model of search prominence with endogenous position value, tested in a lab experiment that varies magnitude of friction (Low vs. High search cost).

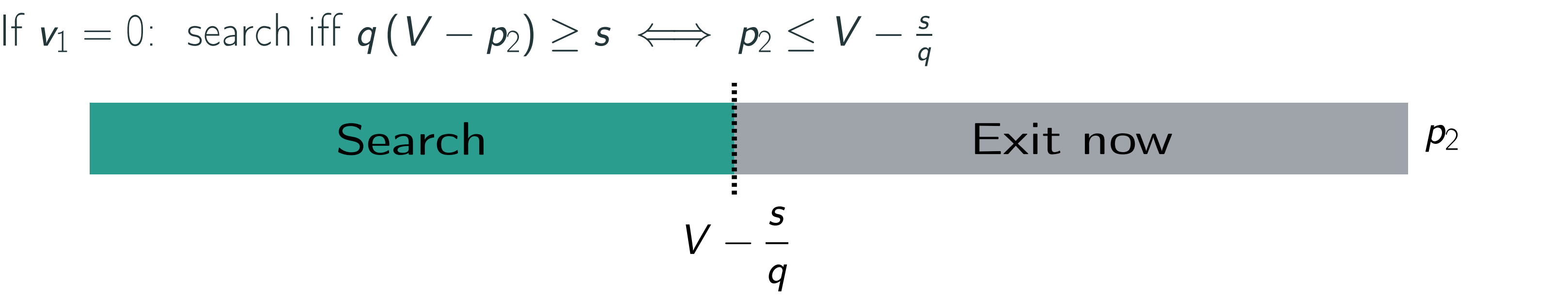
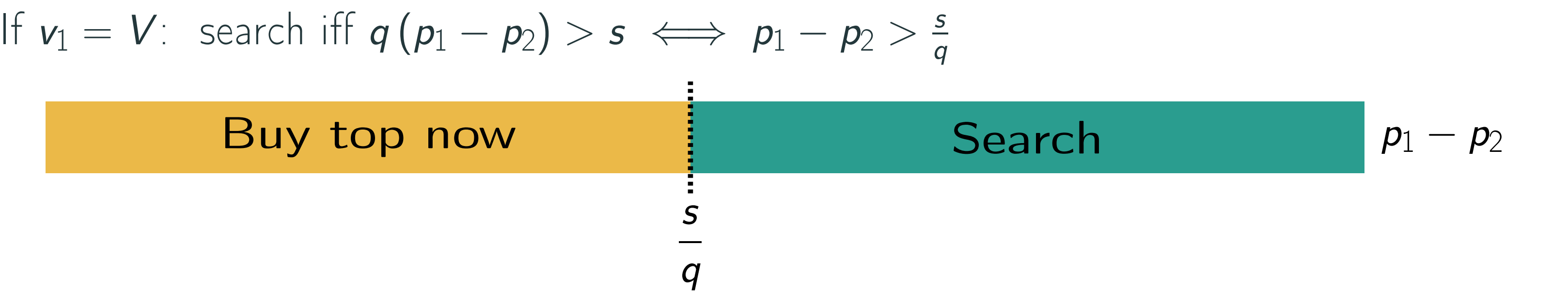
THREE-STAGE GAME



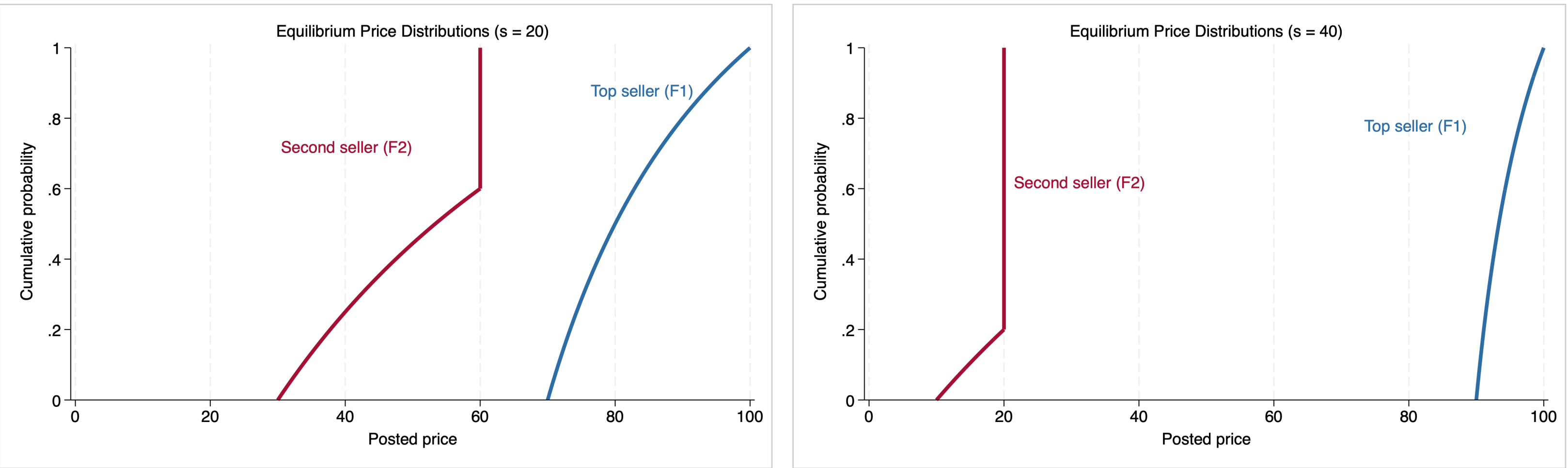
Seller net profit = revenue — **position payment**.
Buyer utility = match value — price — **search cost**.

BUYER'S SEARCH RULE

Given prices (p_1, p_2) and the realized top match $v_1 \in \{0, V\}$, the buyer searches the second seller iff the option value of inspecting exceeds the cost s .



MIXED-STRATEGY PRICING EQUILIBRIUM



- **Price Dispersion:** P_1 and P_2 are dispersed across (disjoint) intervals.
- **Price Hierarchy:** P_1 *first-order stochastically dominates* P_2 .
- **Price Gap:** $(P_1 - P_2) \uparrow$ as $s \uparrow$.

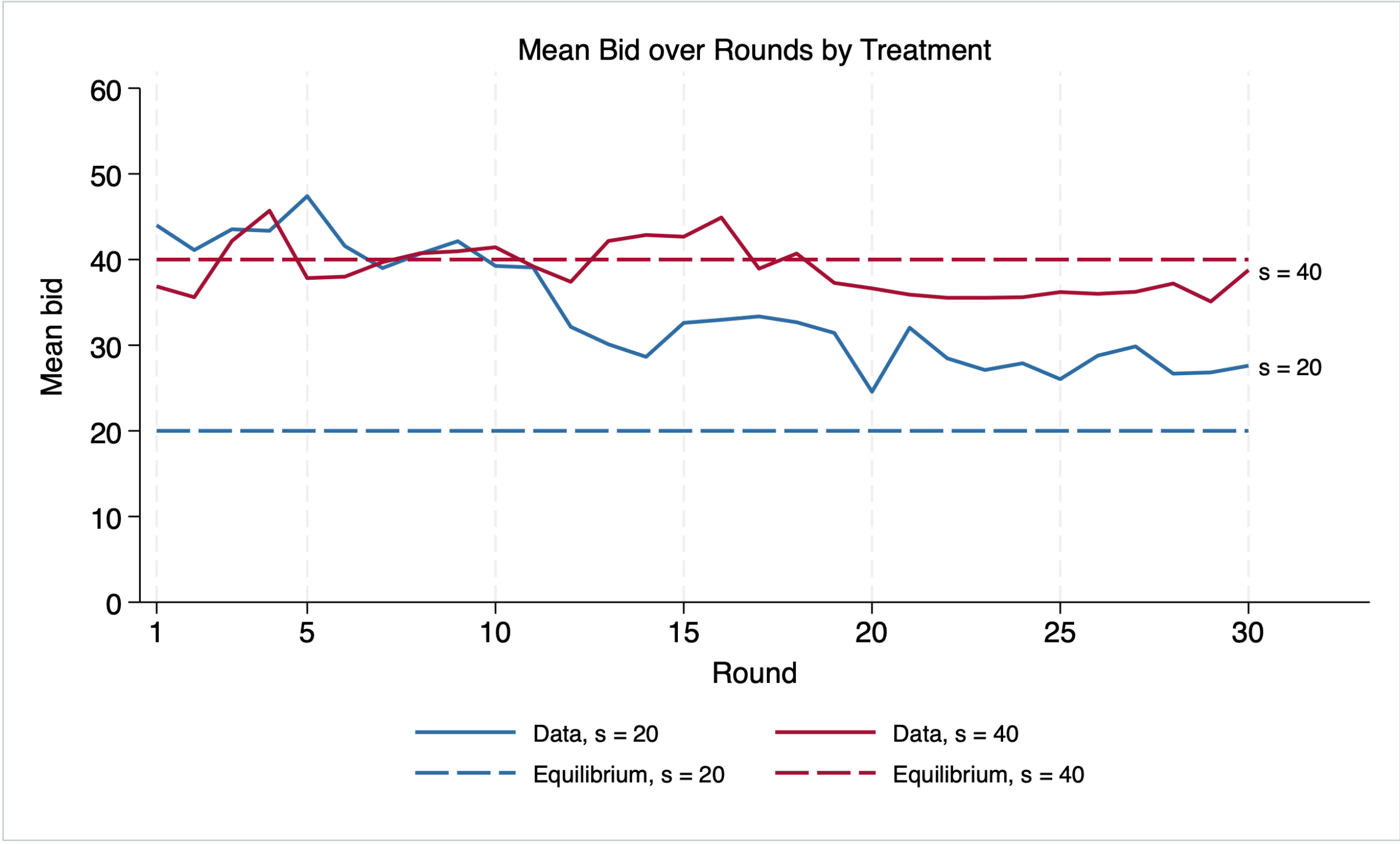
EQUILIBRIUM BIDS AND PAYOFFS

Parameters: $V = 100, q = 0.5$	Low ($s=20$)	High ($s=40$)
Bid $b^* = \Pi_1 - \Pi_2 = s$	20	40
Seller net profit (equal across positions)	15	5
Top price support	[70, 100]	[90, 100]
Second price support	[30, 60]	[10, 20]
Search cutoff in $p_1 - p_2$ ($v_1 = 100$)	40	80
Search cutoff in p_2 ($v_1 = 0$)	60	20

EXPERIMENTAL DESIGN

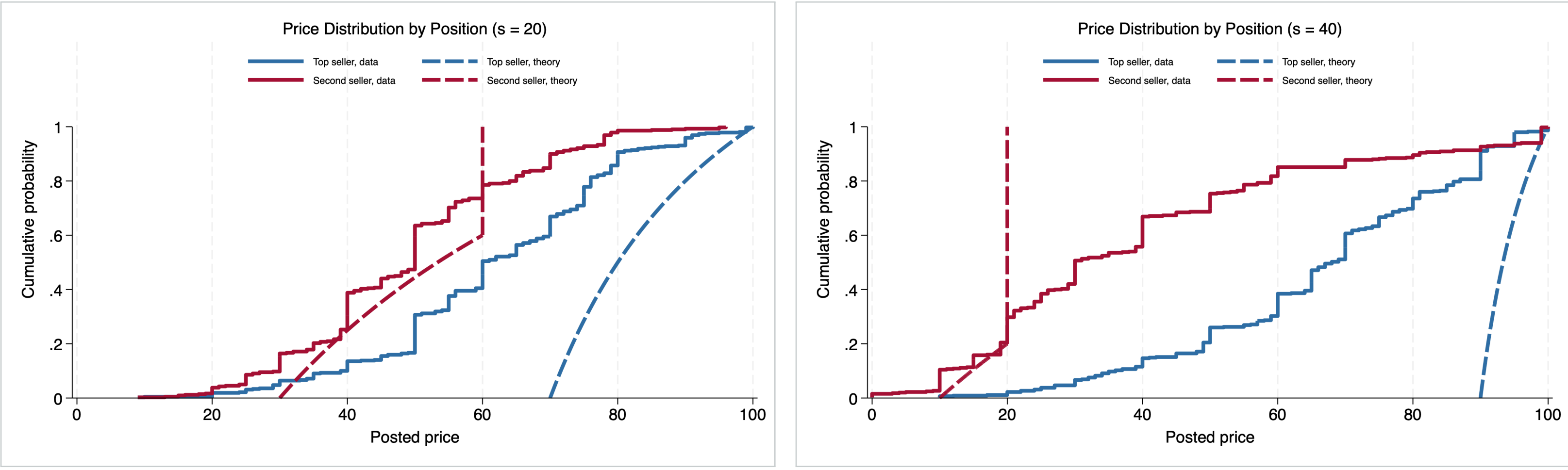
- **Each market:** 2 sellers + 1 buyer, three-stage game. Random re-matching across **30 rounds**, fixed roles; v_1, v_2 drawn in real time.
- **Between-subject:** low search cost ($s=20$) vs. high search cost ($s=40$).
- **Payment:** \$10 show-up + one randomly selected round (10 points = \$1); average \$31.
- **Feedback of on-path history:** prices, top match, second match *if searched*, search decision, purchase, and own payoff; sellers additionally see the position payment.

RESULT 1: SUBMITTED BIDS FOR TOP POSITION



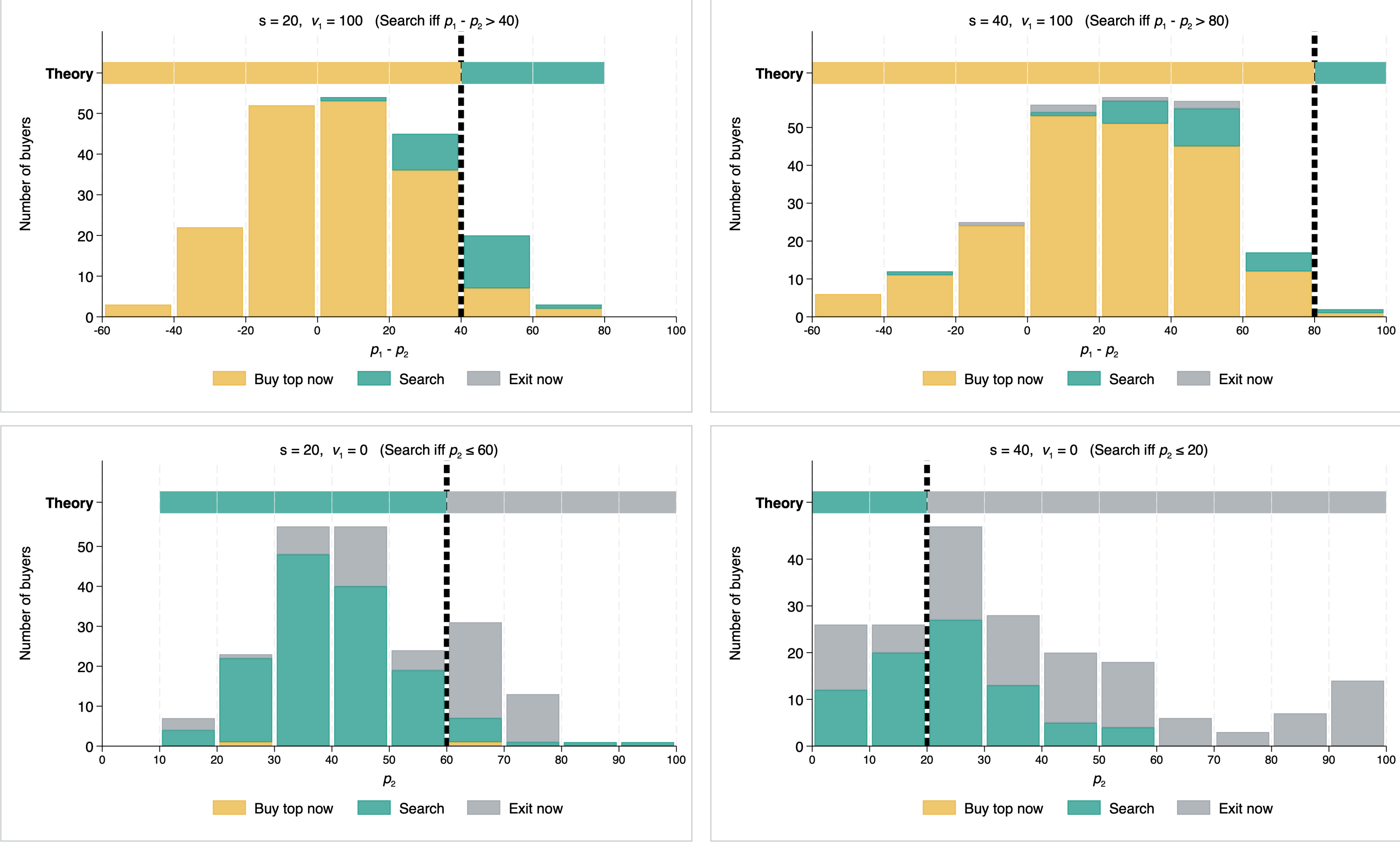
- Overbidding relative to the equilibrium benchmark at $s = 20$; not at $s = 40$.

RESULT 2: PRICE DISPERSION, HIERARCHY, AND GAP



- **Price Dispersion:** wider ranges and higher standard deviation than predicted.
- **Price Hierarchy:** P_1 stochastically dominates P_2 .
- **Price Gap:** mean gap 12.9 ($s=20$) $\rightarrow$ 27.0 ($s=40$), compressed against a predicted 31.7 $\rightarrow$ 74.9.

RESULT 3: SEARCH AND PURCHASE



TAKEAWAYS

- A model of search prominence with endogenous position value and price-directed search.
- Overbidding in low search cost, but not in high search cost.
- Prominence confers market power: $P_1 > P_2$ under both s ; $P_1 - P_2$ increases in s .
- Compressed price gap reshapes search behavior.
- Search patterns broadly consistent with theoretical cutoff strategy. Search errors in both directions, more pronounced upon a bad top match and at $s = 40$.
- Next step: random position allocation as benchmark; welfare and efficiency.